

Competitive Exam Practice 5 — Answer Key

Addressing Modes, CPU Organization, and Bus Structures

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- Q1 (B).** PC-relative addressing computes $EA = [PC] + \text{displacement}$, so the code is position-independent: it can be loaded at any address and still branch correctly, because every reference is relative to the current PC.
- Q2 (D).** PC-relative addressing makes every reference relative to the program counter, so the whole program can be moved (relocated) to a different base address without changing any instruction. Absolute addressing embeds a fixed address; indexed and base-register modes still rely on an absolute base somewhere.
- Q3 (A).** Pointers require **indirect** addressing (the operand is *another address* in memory). With only immediate (constant in the instruction) and direct (absolute address) modes, a pointer's value cannot be fetched and dereferenced. Arrays, records, and recursive procedures can all be built from direct addressing plus constants; pointers cannot.
- Q4 (C).** Indirect addressing $\rightarrow$ pointers (X-2); immediate $\rightarrow$ constants (Y-3); autodecrement $\rightarrow$ loops, since the index register auto-updates as the loop advances (Z-1).
- Q5 (B).** Absolute (direct) addressing puts the *address* of the operand inside the instruction: $EA = A$. (A) describes immediate; (C) describes register-indirect; (D) describes register or stack addressing.
- Q6 (B).** Based and relative addressing compute EA from a register (base or PC) plus a displacement, so the code can be relocated at run time by changing the base register — no code change needed. Absolute addressing is fixed at link time.
- Q7 (B).** A single-bus design must serialize: drive $[R2]$ to one ALU input, then $[R3]$ to the other, then the result back — hence three steps plus two hidden buffers. Three buses (Bus A/B for the operands, Bus C for the result) can carry both operands and the result in the same cycle because they are separate physical paths, so the register-transfer completes in one control step.